

$$A22) \frac{x+2}{x^2-9}$$

$$x^2-9=0$$

$$(x+3)(x-3)=0$$

$$x=3 \text{ or } x=-3$$

Answer: $x \neq 3, x \neq -3$

P29

$$B11) \frac{x^2-4}{x^2-x-2} \cdot \frac{x^2-1}{x+2}$$

$$x^2-x-2 = (x-2)(x+1)$$

$$\Rightarrow x-2 \neq 0 \quad x+1 \neq 0 \quad x+2 \neq 0$$

$$x \neq 2 \quad x \neq -1 \quad x \neq -2$$

$$x^2-4 = (x-2)(x+2)$$

$$x^2-1 = (x-1)(x+1)$$

$$\frac{(x-2)(x+2)}{(x-2)(x+1)} \cdot \frac{(x-1)(x+1)}{x+2} = x-1$$

Restrictions: $x \neq 2, x \neq -1, x \neq -2$

Answer: $x-1$

$$B12) \frac{2}{x-1} - \frac{x+1}{x^2-1}$$

$$x^2-1 = (x-1)(x+1)$$

$$x-1 \neq 0 \Rightarrow x \neq 1$$

$$x+1 \neq 0 \Rightarrow x \neq -1$$

$$C2) x + \frac{1}{x} = 3, x \neq 0$$

$$\left(x + \frac{1}{x}\right)^2 = 3^2$$

$$x^2 + 2 \cdot x \cdot \frac{1}{x} + \left(\frac{1}{x}\right)^2 = 9$$

$$x^2 + \frac{1}{x^2} = 9 - 2 = 7$$

P30

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right) \left(x^2 - x \cdot \frac{1}{x} + \frac{1}{x^2}\right)$$

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} - 1\right)$$

$$x^3 + \frac{1}{x^3} = 3(7-1) = 3 \cdot 6 = 18$$

$$A23) \frac{x^2-9}{x^2+4x+3}$$

Excellent.

$$x^2+4x+3 \neq 0$$

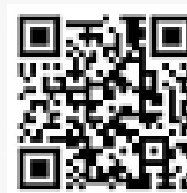
$$x^2+4x+3 = (x+1)(x+3)$$

$$x^2-9 = (x-3)(x+3)$$

$$\frac{(x-3)(x+3)}{(x+1)(x+3)} = \frac{x-3}{x+1}$$

$$\text{Answer: } \frac{x-3}{x+1}$$

Restrictions: $x \neq -1, x \neq -3$



$$\begin{aligned}
 4) \quad & x^4 + 4x^2 + 16 \\
 & x^4 + 4x^2 + 4x^2 + 16 - 4x^2 \\
 & (x^4 + 8x^2 + 16) - 4x^2 \\
 & (x^2 + 4)^2 - 4x^2 \\
 & (x^2 + 4)^2 - (2x)^2 \\
 & (x^2 + 4 - 2x)(x^2 + 4 + 2x)
 \end{aligned}$$

$$A4) \quad x^2 - 7x + 10 = 0$$

$$D = b^2 - 4ac = 9 = 3^2$$

$$x = \frac{-b \pm \sqrt{D}}{2a} = \frac{7 \pm 3}{2} = 5; 2$$

$$x = 2 \text{ or } x = 5$$

$$A6) \quad 3x^2 - 6x + 4 = 0$$

$$D = b^2 - 4ac = 36 - 48 = -12$$

$$D < 0 \Rightarrow \text{no roots}$$

P 59

$$B1) \quad (a-1)x = a^2 - 1$$

$$(a-1)x = (a-1)(a+1)$$

$$1) \quad a-1 \neq 0$$

$$x = \frac{(a-1)(a+1)}{a-1}$$

$$x = a+1$$

$$2) \quad a-1=0$$

$$0 \cdot x = 0$$

$$5) \quad x^2 - 2x + 2$$

$$x^2 - 2x + 2 = (x^2 - 2x + 1) + 1$$

$$(x+1)^2 + 1$$

$$(x+1)^2 + 1 \geq 1$$

$$A5) \quad 2x^2 + 3x - 1 = 0$$

$$D = b^2 - 4ac = 9 + 8 = 17 = \sqrt{17}$$

$$x = \frac{-b \pm \sqrt{D}}{2a} = \frac{-3 \pm \sqrt{17}}{4} = \frac{-3 \pm \sqrt{17}}{4}$$

$$A7) \quad x^2 - 13x^2 + 36 = 0$$

$$1) \quad x^2 = t, \quad t \geq 0$$

$$t^2 - 13t + 36 = 0$$

$$t_1 = 4 \quad t_2 = 9$$

$$2) \quad x^2 = 4 \Rightarrow x = \pm \sqrt{4} \Rightarrow x = \pm 2$$

$$x^2 = 9 \Rightarrow x = \pm \sqrt{9} \Rightarrow x = \pm 3$$

Answer: if $a \neq 1, x = a+1$

if $a = 1, x \in \mathbb{R}$

$$B2) \quad x^2 - 4x - 2 = 0$$

$$x_1 + x_2 = 4$$

$$x_1 x_2 = -2$$

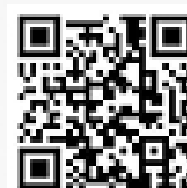
$$x_1^2 x_2 + x_1 x_2^2 = x_1 x_2 (x_1 + x_2)$$

$$B3) \quad x^2 - kx + k = 0 \quad D = 0$$

$$a = 1, \quad b = -k, \quad c = k$$

$$D = b^2 - 4ac = (-k)^2 - 4(1)(k) = k^2 - 4k$$

$$k^2 - 4k = 0$$



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$$(-2) \cdot 4 = -8$$

Answer: -8

$$b4) x^3 - 3x^2 - 4x + 12 = 0$$

$$(x^3 - 3x^2) - (4x - 12) = 0$$

$$x^2(x-3) - 4(x-3) = 0$$

$$(x-3)(x^2-4) = 0$$

$$(x-3)(x-2)(x+2) = 0$$

$$\cdot x-3=0 \Rightarrow x=3$$

$$\cdot x-2=0 \Rightarrow x=2$$

$$\cdot x+2=0 \Rightarrow x=-2$$

Answer: $x = -2, 2, 3$

P59

$$c1) x^2 - (m+1)x + m = 0$$

$$|x_1 - x_2| = 3 \Rightarrow (x_1 - x_2)^2 = 9$$

$$x_1 + x_2 = m+1$$

$$x_1 x_2 = m$$

$$(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1 x_2$$

$$(m+1)^2 - 4m = 9$$

$$m^2 + 2m + 1 - 4m = 9$$

$$m^2 - 2m - 8 = 0$$

$$(m-4)(m+2) = 0$$

Thus, $m = 4$ or $m = -2$

$$k(k-4) = 0$$

Answer: $k = 0, k = 4$

$$b5) (x^2 - 2x)^2 - 5(x^2 - 2x) + 4 = 0$$

$$t^2 - 5t + 4 = 0$$

$$(t-1)(t-4) = 0 \Rightarrow t = 1 \text{ or } t = 4$$

1) if $t = 1$

$$x^2 - 2x - 1 = 0 \Rightarrow x^2 - 2x - 1 = 0$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)} = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

2) if $t = 4$

$$x^2 - 2x = 4 \Rightarrow x^2 - 2x - 4 = 0$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-4)}}{2(1)} = \frac{2 \pm \sqrt{20}}{2} = 1 \pm \sqrt{5}$$

Answer: $x = 1 \pm \sqrt{2}, 1 \pm \sqrt{5}$

$$c5) x + \frac{1}{x} = 4$$

$$1) \left(x + \frac{1}{x}\right)^2 = 4^2$$

$$x^2 + 2\left(x\right)\left(\frac{1}{x}\right) + \frac{1}{x^2} = 16$$

$$x^2 + 2 + \frac{1}{x^2} = 16$$

$$x^2 + \frac{1}{x^2} = 14$$

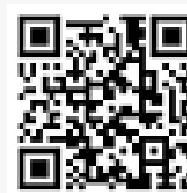
$$2) x^2 + 1 = 4x$$

$$x^2 - 4x + 1 = 0$$

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3}$$

Answer: $x^2 + \frac{1}{x^2} = 14$; $x = 2 \pm \sqrt{3}$



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2026 年 09 月 18 日

Excellent!

P28 A (22) (23), P29 B (11) (12), P30 C (2) (4) (5), P48 2.4.1, P58 A (4) (5) (6) (7),
P59 B (1) (2) (3) (4) (5), P59 C (1) (5)

P28 A (22) (23)

(22). Find the values for which $\frac{n+2}{n^2-9}$ is meaningful

$$\rightarrow n^2 - 9 \neq 0 \leadsto (n-3)(n+3) \neq 0$$

$$n-3 \neq 0$$

$$n \neq 3,,$$

$$n+3 \neq 0$$

$$n \neq -3,,$$

and

$$\rightarrow = \{n | n \neq 3 \text{ and } n \neq -3, n \in \mathbb{R}\},$$

(23) Reduce $\frac{n^2-9}{n^2+4n+3}$, and state the variable restrictions

$$\rightarrow \frac{(n-3)(\cancel{n+3})}{(n+1)(\cancel{n+3})} = \frac{n-3}{n+1} \rightarrow = \frac{n-3}{n+1}, n \neq -1, -3,,$$

$$\hookrightarrow n+1 \neq 0$$

$$n \neq -1,,$$

$$\hookrightarrow n+3 \neq 0$$

$$n \neq -3,,$$

P29 B (11) (12)

(11). Simplify and state all variable restrictions: $\frac{n^2-4}{n^2-n-2} \cdot \frac{n^2-1}{n+2}$

$$\rightarrow \frac{(\cancel{n-2})(\cancel{n+2})}{(\cancel{n+1})(\cancel{n-2})} \cdot \frac{(n-1)(\cancel{n+1})}{(\cancel{n+2})} = (n-1),$$

$$\hookrightarrow n+1 \neq 0$$

$$n \neq -1,,$$

$$\hookrightarrow n-2 \neq 0$$

$$n \neq 2,,$$

$$\hookrightarrow n+2 \neq 0$$

$$n \neq -2,,$$

$$n \neq -1, 2,,$$

$$-1, 2, -2$$

(12). Simplify and state all variable restrictions: $\frac{2}{n-1} - \frac{n+1}{n^2-1} \rightarrow$ faktorkan

$$\rightarrow \frac{2}{n-1} - \frac{\cancel{n+1}}{(\cancel{n-1})(\cancel{n+1})} = \frac{2}{n-1} - \frac{1}{n-1} = \frac{1}{n-1}$$

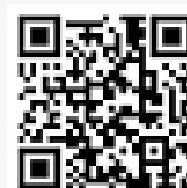
$$\hookrightarrow n-1 \neq 0$$

$$n \neq 1,,$$

$$\hookrightarrow n+1 \neq 0$$

$$n \neq -1,,$$

$$\rightarrow = \frac{1}{n-1}, n \neq -1, 1,,$$



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P 30 C (2)(4)(5)

(2). Given $u + \frac{1}{u} = 3$, with $u \neq 0$, compute $u^2 + \frac{1}{u^2}$, $u^3 + \frac{1}{u^3}$

$\rightarrow u^2 + \frac{1}{u^2} \quad \left\{ \quad u^3 + \frac{1}{u^3} \right\}$ kuadratkan

$$(u + \frac{1}{u})^2 = 3^2 \rightarrow \text{rumus} : a^2 + 2ab + b^2$$

$$u^2 + 2u(\frac{1}{u}) + \frac{1}{u^2} = 9$$

$$u^2 + 2 + \frac{1}{u^2} = 9$$

$$u^2 + \frac{1}{u^2} = 9 - 2 = 7,$$

$$\text{rumus} = a^3 + b^3 = (a+b)^3 - 3ab(a+b)$$

$$a = u, b = \frac{1}{u}$$

$$u^3 + \frac{1}{u^3} = (u + \frac{1}{u})^3 - 3(u \cdot \frac{1}{u})(u + \frac{1}{u})$$

$$= 3^3 - 3 \cdot 1 \cdot 3$$

$$= 27 - 9 = 18,$$

(4). Factor : $u^4 + 4u^2 + 16$

$$\rightarrow = (u^2 + 2u + 4)(u^2 - 2u + 4)$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b)$$

(5). $u^2 + 2u + 2$

$$= u^2 + 2 \cdot u \cdot 1 + 1^2 + 1$$

$$= (u+1)^2 + 1$$

$$= \geq 0 + 1$$

$$= 1, \dots$$

正数 : > 0

负数 : < 0

$$(u^2 + 4)^2 - (2u)^2 \rightarrow (2u)^2$$

$$(u^4 + 8u^2 + 16) - 4u^2$$

$$= u^4 + 4u^2 + 16$$

so,

$$(u^2 - 2u + 4)(u^2 + 2u + 4)$$

P 58 A (4)(5)(6)(7)

(4). Solve $u^2 - 7u + 10 = 0$

$$\rightarrow (u-5)(u-2)$$

$$u - 5 = 0$$

$$u = 5,$$

$$u - 2 = 0$$

$$u = 2,$$

$$u = 2, 5,$$



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A (4) (5) (6) (7)

(5) solve $2x^2 + 3x - 1 = 0$

$\rightarrow a = 2, b = 3, c = -1$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2)(-1)}}{2(2)}$$

$$= \frac{-3 \pm \sqrt{9 + 8}}{4} = \frac{-3 \pm \sqrt{17}}{4}$$

$$\therefore x = \frac{-3 + \sqrt{17}}{4} \text{ or } x = \frac{-3 - \sqrt{17}}{4}$$

(6). Determine the number of real roots of $3x^2 - 6x + 4 = 0$ (akar)

$\rightarrow a = 3, b = -6, c = 4$

$$\Delta = b^2 - 4ac$$

$$= -6^2 - 4(3)(4)$$

$$= 36 - 48 = -12$$

$$\Delta < 0$$

The equation

has 0 real roots.

• If $\Delta > 0$: 2 distinct real roots

• If $\Delta = 0$: 1 repeated real roots

• If $\Delta < 0$: No real roots

(7). Solve $x^4 - 13x^2 + 36 = 0$

$\rightarrow x^4 = (x^2)^2$

$$x^2 - 13x + 36 = 0$$

$$(x-4)(x-9) = 0$$

$$x = 4, x = 9$$

$$\downarrow$$

$$x = \pm 2, x = \pm 3$$

$$x = x^2$$

$$\text{So, } x = -3, -2$$

$$= 2, 3$$

$$\rightarrow x^2 = 4, x^2 = 9$$

P59 B (1) (2) (3) (4) (5)

(1). Discuss the solutions of $(a-1)x = a^2 - 1$

$\rightarrow (a-1)x = a^2 - 1 \rightarrow a^2 = 1$

$$(a-1)(a+1)$$

$$(a-1)x = (a-1)(a+1)$$

① $a \neq 1$

$$\rightarrow \frac{(a-1)x}{(a-1)} = \frac{(a-1)(a+1)}{(a-1)}$$

divide $a-1$

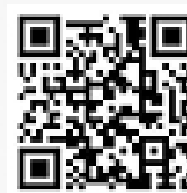
$$(a-1)$$

$$x = a+1$$

② $a = 1 \rightarrow (a-1)x = (a-1)(a+1)$

$$(1-1)x = (1-1)(1+1)$$

$$0x = 0$$



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(2). If u_1, u_2 are the roots of $u^2 - 4u - 2 = 0$, find $u_1^2 u_2 + u_1 u_2^2$

$$\rightarrow \frac{u^2}{a} - \frac{4u}{b} - \frac{2}{c} = 0$$

$$u_1 + u_2 = 4$$

$$u_1 \cdot u_2 = -2$$

$$u_1^2 u_2 + u_1 u_2^2 = u_1 u_2 (u_1 + u_2)$$

$$= (-2)(4)$$

$$= -8$$

韦达定理 $u_1 + u_2 = -\frac{b}{a} = 4$
 $u_1 u_2 = \frac{c}{a} = -2$

(3). The equation $u^2 - ku + k = 0$ in u has two equal real roots, find k .

→ Hint: 2 akar real yg sama $D = 0$

$$D = b^2 - 4ac$$

$$0 = (-k)^2 - 4(1)(k)$$

$$0 = k^2 - 4k$$

$$0 = k(k - 4)$$

$$k_1 = 0$$

$$k - 4 = 0$$

$$k = 4$$

$$k_1 = 0, k_2 = 4$$

(4). Solve $u^3 - 3u^2 - 4u + 12 = 0$

→ try $u = 2$

$$2^3 - 3(2)^2 - 4(2) + 12 = 0$$

$$8 - 3(4) - 8 + 12 = 0$$

$$u - 2 = 0$$

$$u = 2, \text{ factor}$$

$$u^3 - 3u^2 - 4u + 12 = 0$$

$$(u - 2)(u^2 - u - 6)$$

$$(u - 2)(u - 3)(u + 2) = 0$$

$$u = 2, u = 3, u = -2$$

$$u = -2, 2, 3$$

(5). Solve $(u^2 - 2u)^2 - 5(u^2 - 2u) + 4 = 0$ | If: $u = u^2 - 2u$

$$\rightarrow u^2 - 5u + 4 = 0$$

$$(u - 1)(u - 4) = 0$$

$$\rightarrow u = 1, u = 4$$

$$\textcircled{1}. u^2 - 2u = 1 \rightarrow u^2 - 2u - 1 = 0$$

$$u = 1 \pm \sqrt{2}$$

$$\textcircled{2}. u^2 - 2u = 4 \rightarrow u^2 - 2u - 4 = 0$$

$$u = 1 \pm \sqrt{5}$$

$$\text{So, } u = 1 + \sqrt{2}, 1 - \sqrt{2}, 1 + \sqrt{5}, 1 - \sqrt{5}$$



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9 c (1)(5)

(1). If the two roots of $u^2 - (m+1)u + m = 0$ differ by 3, find m .

→ Vieta

If. akar / u_1, u_2

$$u_1 + u_2 = m+1$$

$$u_1 u_2 = m$$

$$au^2 + bu + c = 0$$

$$u^2 - (m+1)u + m = 0$$

differ by 3

$$a = 1$$

$$b = -(m+1)$$

$$c = m$$

$$\left. \begin{array}{l} \text{sum of roots } \alpha + \beta = -\frac{b}{a} = m+1 \\ \text{product of roots } \alpha \cdot \beta = m \end{array} \right\}$$

$$\left. \begin{array}{l} \text{sum of roots } \alpha + \beta = -\frac{b}{a} = m+1 \\ \text{product of roots } \alpha \cdot \beta = m \end{array} \right\}$$

$$|u_1 - u_2| = 3 \rightarrow |\alpha - \beta| = 3$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$3^2 = (m+1)^2 - 4m$$

$$3^2 = m^2 + 2m + 1 - 4m$$

$$9 = m^2 - 2m + 1$$

$$0 = m^2 - 2m - 8$$

$$(m-4)(m+2) = 0 \rightarrow m = 4, m = -2$$

(5). Given $u + 1/u = 4$, find $u^2 + 1/u^2$, and solve for all possible u

→ $u + \frac{1}{u} = 4$, find $u^2 + \frac{1}{u^2}$, semua kemungkinan u

$$(u^2 + \frac{1}{u^2}) = ? \sim (u + \frac{1}{u})^2 = 4^2$$

$$u^2 + 2(u)(\frac{1}{u}) + \frac{1}{u^2} = 16$$

$$u^2 + \frac{1}{u^2} + 2 = 16$$

$$u^2 + \frac{1}{u^2} = 16 - 2$$

$$u^2 + \frac{1}{u^2} = 14$$

$$u + \frac{1}{u} = 4$$

_____ kali u

$$u^2 + 1 = 4u$$

$$u^2 - 4u + 1 = 0$$

$$u = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

$$\text{So, } u = 2 + \sqrt{3} \text{ or } u = 2 - \sqrt{3}$$



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P48 2.4.1

(1). Solve $x^3 - 4x = 0$

$\rightarrow x(x^2 - 4) = 0$

$x(x-2)(x+2) = 0$

$x = 0$

$x-2=0$

$x+2=0$

$x=2$

$x=-2$

$\rightarrow x = -2 \text{ 或 } 2 \text{ 或 } 0$

(2). $x^3 + 2x^2 - x - 2 = 0$

$\rightarrow (x^3 + 2x^2) - (x + 2) = 0$

$x^2(x+2) - (x+2) = 0$

$(x+2)(x^2-1) = 0$

$(x+2)(x-1)(x+1) = 0$

$x+2=0$

$x=-2$

$x-1=0$

$x=1$

$x+1=0$

$x=-1$

$\rightarrow x = \{-2, -1, 1\}$ $x = -2, x = 1 \text{ 或 } -1$

(3). Solve $x^4 - 10x^2 + 9 = 0$

$\rightarrow \text{Let } x^2 = p$

$(x^2)^2 - 10x^2 + 9 = 0$

$p^2 - 10p + 9 = 0$

$(p-1)(p-9) = 0$

$p-1=0$

$p=1$

$p = x^2$

$x^2 - 1 = 0$

$(x-1)(x+1) = 0$

$x-1=0$

$x=1$

$x+1=0$

$x=-1$

$p = x^2$

$x^2 = 9$

$x^2 - 9 = 0$

$(x-3)(x+3) = 0$

$x-3=0$ $\left\{ \begin{array}{l} x+3=0 \\ x=3 \end{array} \right.$

$x=3$ $\left\{ \begin{array}{l} x=-3 \end{array} \right.$

$\rightarrow x = \{-3, -1, 1, 3\} \rightarrow \boxed{x = 1 \text{ 或 } -1, x = 3 \text{ 或 } -3}$

(4). Solve $(x-1)^4 - 5(x-1)^2 + 4 = 0$

$\rightarrow \text{Let } p = (x-1)^2$

$((x-1)^2)^2 - 5(x-1)^2 + 4 = 0$

$p^2 - 5p + 4 = 0$

$(p-1)(p-4) = 0$

$p-1=0$

$p=1$

$p-4=0$

$p=4$

$p=1 \text{ 或 } p=4$

$p = (x-1)^2$

$(x-1)^2 = 1$

$x-1 = \pm\sqrt{1}$

$x-1 = \pm 1$

$(x-1)^2 = 4$

$x-1 = \pm\sqrt{4}$

$x-1 = \pm 2$

$x = 1+2$

$x = 3$

$x = 1-2$

$x = -1$

$x = 1+1$

$x = 2$

$x = 1-1$

$x = 0$

$x = \{-1, 0, 2, 3\}$



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P28 (A)

Excellent

$$\textcircled{22} \quad \frac{x+2}{x^2-9} \Rightarrow$$

$$\Rightarrow \frac{x+2}{(x+3)(x-3)} \Rightarrow$$

$$\Rightarrow (x+3)(x-3) \neq 0$$

$$1. \quad x+3 \neq 0$$

$$x \neq -3$$

$$2. \quad x-3 \neq 0$$

$$x \neq 3$$

$$x \in (-\infty; -3) \cup (-3; 3) \cup (3; +\infty)$$

$$\textcircled{23} \quad \frac{x^2-9}{x^2+4x+3} \Rightarrow$$

$$\Rightarrow \frac{(x+3)(x-3)}{(x+3)(x+1)} \Rightarrow$$

$$\Rightarrow \frac{(x-3)}{(x+1)}$$

$$1. \quad x+3 \neq 0$$

$$x \neq -3$$

$$2. \quad x+1 \neq 0$$

$$x \neq -1$$

$$x \in (-\infty; -3) \cup (-3; -1) \cup (-1; +\infty)$$

P29 (B)

$$\textcircled{11} \quad \frac{x^2-4}{x^2-x-2} \times \frac{x^2-1}{x+2} \Rightarrow$$

$$\Rightarrow \frac{(x+2)(x-2)}{(x-2)(x+1)} \times \frac{(x+1)(x-1)}{(x+2)} \Rightarrow$$

$$\Rightarrow \frac{(x+2)(x-2)(x+1)(x-1)}{(x-2)(x+1)(x+2)} \Rightarrow$$

$$\Rightarrow x-1=0$$

$$\Rightarrow x=1 \quad x \neq 2, -2, -1$$

$$\textcircled{12} \quad \frac{2}{x-1} - \frac{x+1}{x^2-1} \Rightarrow$$

$$\Rightarrow \frac{2}{x-1} - \frac{x+1}{(x+1)(x-1)} \Rightarrow$$

$$\Rightarrow \frac{2(x+1) - (x+1)}{(x+1)(x-1)} \Rightarrow$$

$$\Rightarrow \frac{2x+2-x-1}{(x+1)(x-1)} \Rightarrow$$

$$\Rightarrow \frac{x+1}{(x+1)(x-1)} \Rightarrow \frac{1}{x-1} = 0$$

$$\frac{1}{x-1} = 0 \Rightarrow x-1 \neq 0 \Rightarrow x \neq 1$$

$$x \in (-\infty; 1) \cup (1; +\infty)$$

$$x \neq 1, x \neq -1$$

P30 (C)

$$\textcircled{2} \quad 1. \quad x^2 + \frac{1}{x^2} \Rightarrow$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 3^2 \Rightarrow$$

$$\Rightarrow x^2 + 2 \cdot x \cdot \frac{1}{x} + \frac{1}{x^2} = 9 \Rightarrow$$

$$\Rightarrow x \cdot \frac{1}{x} = 1 \Rightarrow x^2 + 2 + \frac{1}{x^2} = 9 \Rightarrow$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 9 - 2 = 7$$

$$2. \quad x^3 + \frac{1}{x^3} \Rightarrow$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right) \left(x^2 - x \cdot \frac{1}{x} + \frac{1}{x^2}\right) \Rightarrow$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right) \left(x^2 + \frac{1}{x^2} - 1\right) \Rightarrow$$

$$\Rightarrow \left(x + \frac{1}{x}\right) = 3 \quad \text{and} \quad \left(x^2 + \frac{1}{x^2}\right) = 7 \Rightarrow$$

$$\Rightarrow x^3 + \frac{1}{x^3} = 3 \cdot (7 - 1) = 3 \cdot 6 = 18$$

$$\textcircled{4} \quad x^4 - 4x^2 + 16 = (x^2 - 2x + 4)(x^2 + 2x + 4)$$

$$x^4 = (x^2)^2 \quad (x^2 + 4)^2 = 2 \cdot x^2 \cdot 4 + 8x^2 \Rightarrow$$

$$16 = 4^2$$

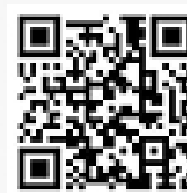
$$\Rightarrow (x^4 + 8x^2 + 16) - 4x^2 \Rightarrow (a+b)^2 = a^2 + 2ab + b^2$$

$$\Rightarrow (x^2 + 4)^2 - 4x^2 \Rightarrow (x^2 + 4)^2 - (2x)^2 \Rightarrow$$

$$\Rightarrow (a-b)^2 = (a-b)(a+b) \Rightarrow$$

$$\Rightarrow ((x^2 + 4) - 2x)((x^2 + 4) + 2x) \Rightarrow$$

$$\Rightarrow (x^2 - 2x + 4)(x^2 + 2x + 4)$$



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$$⑤ (a+b)^2 = a^2 + 2ab + b^2$$

$$x^2 + 2x + 2 = x^2 + 2x + 1 + 1 \Rightarrow$$

$$\Rightarrow (x^2 + 2x + 1) + 1 = (x+1)^2 + 1 \Rightarrow$$

$$\Rightarrow (x+1)^2 + 1 \geq 0 + 1$$

$$\Rightarrow (x+1)^2 + 1 \geq 1$$

在实数范围内,任何数的平方都大于或等于0.

一个非负数加上1,其结果必然恒大于0.

P48 2.4.1

$$① x^3 - 4x = 0$$

$$\Rightarrow x(x^2 - 4) = 0$$

$$\Rightarrow x(x-2)(x+2) = 0$$

$$x_1 = 0$$

$$x_2 = 2$$

$$x_3 = -2$$

$$② x^3 + 2x^2 - x - 2 = 0$$

$$\Rightarrow (x^3 + 2x^2) - (x + 2) = 0$$

$$\Rightarrow x^2(x+2) - (x+2) = 0$$

$$\Rightarrow (x+2)(x^2 - 1) = 0$$

$$\Rightarrow (x+2)(x-1)(x+1) = 0$$

$$x_1 = -2$$

$$x_2 = 1$$

$$x_3 = -1$$

$$③ x^4 - 10x^2 + 9 = 0$$

$$\Rightarrow t = x^2$$

$$\Rightarrow t^2 - 10t + 9 = 0$$

$$\Rightarrow t_1 + t_2 = 10$$

$$t_1 \cdot t_2 = 9$$

$$\Rightarrow t_1 = 1$$

$$t_2 = 9$$

$$\Rightarrow x_{1,2} = \pm 1 \quad [x^2 = 1]$$

$$x_{3,4} = \pm 3 \quad [x^2 = 9]$$

$$④ (x-1)^4 - 5(x-1)^2 + 4 = 0$$

$$\Rightarrow (x-1)^2 = t$$

$$\Rightarrow t^2 - 5t + 4 = 0$$

$$\Rightarrow t_1 + t_2 = 5; t_1 \cdot t_2 = 4$$

$$\Rightarrow t_1 = 1; t_2 = 4$$

$$\Rightarrow (x-1)^2 = 1; (x-1)^2 = 4$$

$$1. (x-1)^2 = 1 \Rightarrow x-1 = 1 \Rightarrow x_1 = 2$$

$$2. (x-1)^2 = 1 \Rightarrow x-1 = -1 \Rightarrow x_2 = 0$$

$$3. (x-1)^2 = 4 \Rightarrow x-1 = 2 \Rightarrow x_3 = 3$$

$$(x-1)^2 = 4 \Rightarrow x-1 = -2 \Rightarrow x_4 = -1$$

P58 (A)

$$④ x^2 - 7x + 10 = 0$$

$$\Rightarrow x_{1,2} = \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \cdot 1 \cdot 10}}{2 \cdot 1} = \frac{7 \pm 3}{2}$$

$$\Rightarrow x_1 = \frac{7+3}{2} = \frac{10}{2} = 5$$

$$\Rightarrow x_2 = \frac{7-3}{2} = \frac{4}{2} = 2$$

$$⑤ 2x^2 + 3x - 1 = 0$$

$$\Rightarrow x_{1,2} = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 2 \cdot (-1)}}{2 \cdot 2} = \frac{-3 \pm \sqrt{17}}{4}$$

$$\Rightarrow x_1 = \frac{-3 + \sqrt{17}}{4}; x_2 = \frac{-3 - \sqrt{17}}{4}$$

$$⑥ 3x^2 - 6x + 4 = 0$$

$$\Rightarrow D = b^2 - 4ac \Rightarrow D = (-6)^2 - 4 \cdot 3 \cdot 4 = -12$$

$$D < 0$$

因为判别式小于0 ($D < 0$), 所以在实数范围内无解.

$$⑦ x^4 - 13x^2 + 36 = 0 \Rightarrow x^2 = t$$

$$\Rightarrow t^2 - 13t + 36 = 0$$

$$\Rightarrow t_1 + t_2 = 13; t_1 \cdot t_2 = 36$$

$$\Rightarrow t_1 = 4 \Rightarrow x^2 = 4 \Rightarrow x_1 = 2; x_2 = -2$$

$$\Rightarrow t_2 = 9 \Rightarrow x^2 = 9 \Rightarrow x_3 = 3; x_4 = -3$$



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P59 (B)

$$(1) (a-1)x = a^2 - 1$$

$$\Rightarrow (a-1)x = (a-1)(a+1)$$

1. 当 $a=1$ 时: 方程变为 $0x=0$; 有无数个解 (x 为任意实数).

2. 当 $a=-1$ 时: 方程变为 $-2x=0$, 有唯一解: $x=0$

3. 当 $a \neq \pm 1$ 时: 两边约去 $(a-1)$, 有唯一解: $x=a+1$

$$(2) x^2 - 4x - 2 = 0 \Rightarrow$$

$$x_1^2 x_2 + x_1 x_2^2 = x_1 x_2 (x_1 + x_2)$$

$$\Rightarrow x_1 + x_2 = -(-4) = 4$$

$$x_1 \cdot x_2 = -2$$

$$\Rightarrow x_1 + x_2 \cdot x_1 \cdot x_2 (x_1 + x_2) = -2 \cdot 4 = -8$$

$$(3) x^2 - kx + k = 0$$

$$D = b^2 - 4ac$$

$$\Rightarrow D = (-k)^2 - 4 \cdot 1 \cdot k \Rightarrow$$

$$\Rightarrow D = k^2 - 4k$$

$$\Rightarrow k(k-4) = 0$$

$$k_1 = 0 \Rightarrow x^2 - 0x + 4 = 0 \Rightarrow (x-2)^2 = 0 \Rightarrow x_1 = x_2 = 2$$

$$k_2 = 4 \Rightarrow x^2 = 0; x_1 = x_2 = 0$$

$$(4) x^3 - 3x^2 - 4x + 12 = 0$$

$$\Rightarrow (x^3 - 3x^2) - (4x - 12) = 0$$

$$\Rightarrow x^2(x-3) - 4(x-3) = 0$$

$$\Rightarrow (x-3)(x^2-4) = 0$$

$$\Rightarrow (x-3)(x+2)(x-2) = 0$$

$$x_1 = 3$$

$$x_2 = -2$$

$$x_3 = 2$$

$$(5) (x^2 - 2x)^2 - 5(x^2 - 2x) + 4 = 0$$

$$x^2 - 2x = t$$

$$\Rightarrow t^2 - 5t + 4 = 0$$

$$t_1 + t_2 = 5$$

$$t_1 \cdot t_2 = 4$$

$$\Rightarrow t_1 = 1; t_2 = 4$$

$$\Rightarrow x^2 - 2x - 1 = 0 \Rightarrow x^2 - 2x - 1 = 0$$

$$x_{1,2} = \frac{2 \pm \sqrt{4}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = 1 \pm \sqrt{2}$$

$$\Rightarrow t = 4$$

$$\Rightarrow x^2 - 2x = 4 \Rightarrow x^2 - 2x - 4 = 0$$

$$x_{3,4} = \frac{2 \pm \sqrt{20}}{2} = \frac{2 \pm 2\sqrt{5}}{2} = 1 \pm \sqrt{5}$$

P59 (C)

$$(1) x^2 - (m+1)x + m = 0$$

$$|x_1 - x_2| = \frac{\sqrt{D}}{|a|}$$

$$\sqrt{D} = 3, \Rightarrow D = 9$$

$$\Rightarrow D = b^2 - 4ac = (m+1)^2 - 4 \cdot m \cdot 1$$

$$\Rightarrow D = m^2 + 2m + 1 - 4m = m^2 - 2m + 1$$

$$\Rightarrow D = (m-1)^2$$

$$\Rightarrow (m-1)^2 = 9$$

$$\Rightarrow m-1 = 3 \Rightarrow m = 4$$

$$\Rightarrow m-1 = -3 \Rightarrow m = -2$$

如果 $m = 4$: ~~$x^2 - 5x + 4 = 0$~~

$$x^2 - 5x + 4 = 0 \Rightarrow$$

$$\Rightarrow (x-4)(x-1) = 0$$

如果 $m = -2$: $x^2 + x - 2 = 0$

$$(5) x^2 + \frac{1}{x^2}; \left(x + \frac{1}{x}\right)^2 = 4^2$$

$$\Rightarrow (a+b)^2 = a^2 + 2ab + b^2$$

$$\Rightarrow x^2 + 2 \cdot x \cdot \frac{1}{x} + \frac{1}{x^2} = 16$$

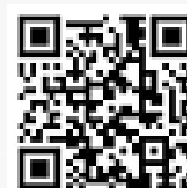
$$x \cdot \frac{1}{x} = 1$$

$$\Rightarrow x^2 + 2 + \frac{1}{x^2} = 16$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 16 - 2 = 14$$

$$x + \frac{1}{x} = 4$$

$$\Rightarrow x^2 + 1 = 4x \Rightarrow x^2 - 4x + 1 = 0$$



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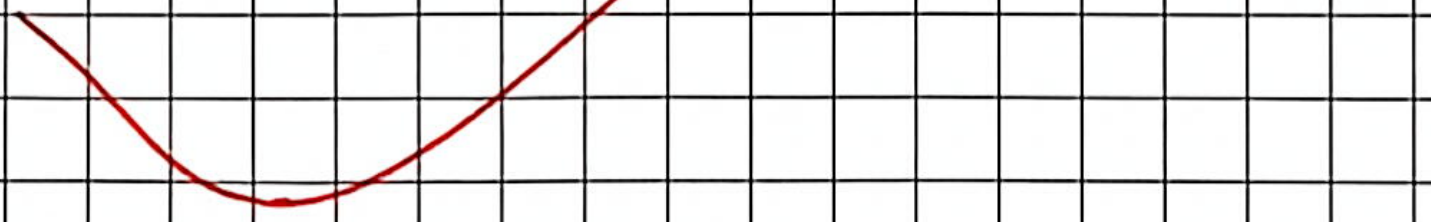
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$$D = b^2 - 4ac = (-4)^2 - 4 \cdot 1 \cdot 1 = 16 - 4 = 12$$

$$\Rightarrow x_{1,2} = \frac{4 \pm \sqrt{12}}{2} \quad [\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}]$$

$$\Rightarrow x_{1,2} = \frac{4 \pm 2\sqrt{3}}{2}$$

$$\Rightarrow x = 2 \pm \sqrt{3}$$



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